

Hospital Patient Flow System

Database Schema Documentation

Version 1.0 | March 2026 | Multi-Tenant SaaS Platform

Overview

This document defines the complete database schema for the Hospital Patient Flow System. The system uses a multi-tenant SaaS architecture with two separate database layers:

Database	Tables	Purpose
Tenant DB (per hospital)	32 tables	All clinical operations patients, visits, triage, consultations, lab, radiology, pharmacy, billing, wards, admin. One isolated copy per subscribing hospital.
Master DB (system owner)	8 tables	Business layer tenant registry, subscriptions, invoices, SaaS payments, super admin accounts, and audit logs. One shared central database.

Column flag legend:

Flag	Meaning
PK	Primary Key — unique identifier for each row
FK	Foreign Key — references a row in another table (shown as → table.column)
UQ	Unique — value must be unique across all rows in the table
NN	Not Null — field is required and cannot be empty

Part 1 Tenant Database (Per Hospital)

Every subscribing hospital has its own isolated copy of this database. No data is shared between hospitals. The tables below exist independently for each hospital tenant.

Reception

patients

Master patient registry. One record per unique patient.

Column	Data Type	Flags	Description
patient_id	UUID	PK	Unique patient identifier
patient_number	VARCHAR(20)	UQ NN	Auto-generated unique patient number
full_name	VARCHAR(200)	NN	Patient full legal name
date_of_birth	DATE	NN	Date of birth
gender	ENUM	NN	male / female / other
national_id	VARCHAR(50)	UQ	National ID or passport number
phone_primary	VARCHAR(20)	NN	Primary contact phone
phone_secondary	VARCHAR(20)		Secondary contact phone
email	VARCHAR(150)		Email address
address	TEXT		Residential address
next_of_kin_name	VARCHAR(200)		Next of kin full name
next_of_kin_phone	VARCHAR(20)		Next of kin contact number
next_of_kin_relationship	VARCHAR(50)		Relationship to patient
blood_group	VARCHAR(5)		Blood group e.g. A+, O-
allergies	TEXT		Known allergies (free text)
is_active	BOOLEAN	NN	Whether patient record is active
created_by	UUID	FK	→ users.user_id — receptionist who registered
created_at	TIMESTAMP	NN	Registration timestamp
updated_at	TIMESTAMP	NN	Last update timestamp

visits

One record per patient attendance/encounter. Links to all clinical activity for that visit.

Column	Data Type	Flags	Description
visit_id	UUID	PK	Unique visit identifier
patient_id	UUID	FK NN	→ patients.patient_id
visit_number	VARCHAR(20)	UQ NN	Auto-generated visit number
visit_date	DATE	NN	Date of the visit
visit_type	ENUM	NN	outpatient / inpatient / emergency
payment_type	ENUM	NN	cash / insurance
insurance_id	UUID	FK	→ patient_insurance.insurance_id
queue_number	VARCHAR(10)		Queue number assigned at reception
status	ENUM	NN	registered → triaged → in_consultation → ... → completed
registered_by	UUID	FK NN	→ users.user_id — receptionist
created_at	TIMESTAMP	NN	Visit creation timestamp
updated_at	TIMESTAMP	NN	Last status update timestamp

patient_insurance

Insurance policy details per patient. One patient may have multiple policies over time.

Column	Data Type	Flags	Description
insurance_id	UUID	PK	Unique insurance record ID
patient_id	UUID	FK NN	→ patients.patient_id
insurer_name	VARCHAR(150)	NN	Name of the insurance company
policy_number	VARCHAR(100)	NN	Patient policy number
coverage_limit	DECIMAL(12,2)		Maximum coverage amount
expiry_date	DATE		Policy expiry date
verification_status	ENUM	NN	pending / verified / rejected
verified_at	TIMESTAMP		Timestamp of verification
is_active	BOOLEAN	NN	Whether this insurance record is active
created_at	TIMESTAMP	NN	Record creation timestamp

queues

Tracks patient position and status in each department queue throughout a visit.

Column	Data Type	Flags	Description
queue_id	UUID	PK	Unique queue entry ID
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
queue_type	ENUM	NN	triage / doctor / lab / radiology / pharmacy / billing
queue_number	VARCHAR(10)	NN	Display queue number
priority	ENUM	NN	emergency / urgent / semi_urgent / non_urgent
status	ENUM	NN	waiting / in_progress / completed / skipped
called_at	TIMESTAMP		When patient was called
completed_at	TIMESTAMP		When service was completed
created_at	TIMESTAMP	NN	Queue entry creation time

Triage

triage_assessments

Vital signs and triage category assigned by nurse. One record per visit.

Column	Data Type	Flags	Description
triage_id	UUID	PK	Unique triage assessment ID
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
triage_nurse_id	UUID	FK NN	→ users.user_id — nurse who triaged
blood_pressure_systolic	INT		Systolic BP in mmHg
blood_pressure_diastolic	INT		Diastolic BP in mmHg
temperature	DECIMAL(4,1)		Temperature in °C
pulse_rate	INT		Pulse rate in bpm
oxygen_saturation	DECIMAL(4,1)		SpO2 percentage
respiratory_rate	INT		Breaths per minute
weight_kg	DECIMAL(5,1)		Weight in kilograms
chief_complaint	TEXT	NN	Patient presenting complaint (free text)
complaint_code	VARCHAR(20)		Structured complaint code if selected from list
triage_category	ENUM	NN	emergency / urgent / semi_urgent / non_urgent
triage_notes	TEXT		Additional nurse notes
assessed_at	TIMESTAMP	NN	Time triage was completed

Doctor Consultation

consultations

Doctor's clinical record for a visit. One record per visit per doctor.

Column	Data Type	Flags	Description
consultation_id	UUID	PK	Unique consultation ID
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
doctor_id	UUID	FK NN	→ users.user_id — attending doctor
presenting_history	TEXT		History of presenting illness
examination_findings	TEXT		Clinical examination findings
clinical_impression	TEXT		Doctor's clinical impression
disposition	ENUM		outpatient / admission / referral / deceased
referral_type	ENUM		internal / external
referral_notes	TEXT		Referral details and destination
consultation_status	ENUM	NN	in_progress / completed
started_at	TIMESTAMP	NN	Consultation start time
completed_at	TIMESTAMP		Consultation completion time

diagnoses

Stores provisional, differential, and final diagnoses separately. Multiple rows per consultation.

Column	Data Type	Flags	Description
diagnosis_id	UUID	PK	Unique diagnosis record ID
consultation_id	UUID	FK NN	→ consultations.consultation_id
patient_id	UUID	FK NN	→ patients.patient_id
diagnosis_type	ENUM	NN	provisional / differential / final
icd10_code	VARCHAR(10)		ICD-10 diagnostic code
diagnosis_text	TEXT	NN	Diagnosis description
sequence_order	INT		Order for differential diagnoses
recorded_by	UUID	FK NN	→ users.user_id — doctor
recorded_at	TIMESTAMP	NN	Timestamp of diagnosis entry

investigation_requests

Doctor's requests for lab tests or radiology. Shared table used by both modules.

Column	Data Type	Flags	Description
request_id	UUID	PK	Unique investigation request ID
visit_id	UUID	FK NN	→ visits.visit_id
consultation_id	UUID	FK NN	→ consultations.consultation_id
patient_id	UUID	FK NN	→ patients.patient_id
request_type	ENUM	NN	lab / radiology
test_name	VARCHAR(200)	NN	Name of test or imaging study
test_code	VARCHAR(50)		Internal test code
clinical_indication	TEXT	NN	Clinical reason for the request
urgency	ENUM	NN	routine / urgent / stat
requested_by	UUID	FK NN	→ users.user_id — doctor
status	ENUM	NN	pending / specimen_collected / in_progress / completed / cancelled
requested_at	TIMESTAMP	NN	Request creation timestamp

prescriptions

Electronic prescriptions issued by doctors. Each row is one drug item.

Column	Data Type	Flags	Description
prescription_id	UUID	PK	Unique prescription ID
visit_id	UUID	FK NN	→ visits.visit_id
consultation_id	UUID	FK NN	→ consultations.consultation_id
patient_id	UUID	FK NN	→ patients.patient_id
drug_name	VARCHAR(200)	NN	Generic or brand drug name
dose	VARCHAR(50)	NN	Dose amount e.g. 500mg
frequency	VARCHAR(50)	NN	Frequency e.g. twice daily
duration	VARCHAR(50)	NN	Duration e.g. 7 days
route	ENUM	NN	oral / iv / im / topical / inhaled / sublingual / other
instructions	TEXT		Additional dispensing instructions
prescribed_by	UUID	FK NN	→ users.user_id — doctor
status	ENUM	NN	pending / dispensed / cancelled
prescribed_at	TIMESTAMP	NN	Prescription creation timestamp

Laboratory

lab_results

Test results entered by lab technicians. One row per test result.

Column	Data Type	Flags	Description
result_id	UUID	PK	Unique lab result ID
request_id	UUID	FK NN	→ investigation_requests.request_id
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
specimen_type	VARCHAR(100)		Type of specimen e.g. blood, urine
specimen_collected_at	TIMESTAMP		Specimen collection timestamp
specimen_label	VARCHAR(50)		Specimen barcode or label ID
result_value	TEXT		Numeric or categorical result value
unit	VARCHAR(30)		Unit of measurement
reference_range	VARCHAR(100)		Normal reference range for the test
is_critical	BOOLEAN	NN	Whether result is a critical/panic value
critical_notified_at	TIMESTAMP		Timestamp critical value was notified to doctor
result_notes	TEXT		Lab technician notes
performed_by	UUID	FK NN	→ users.user_id — lab technician
verified_by	UUID	FK	→ users.user_id — supervisor
status	ENUM	NN	pending / in_progress / resulted / verified
resulted_at	TIMESTAMP		Timestamp result was entered

specimens

Physical specimen tracking from collection to processing.

Column	Data Type	Flags	Description
specimen_id	UUID	PK	Unique specimen ID
request_id	UUID	FK NN	→ investigation_requests.request_id
patient_id	UUID	FK NN	→ patients.patient_id
specimen_type	VARCHAR(100)	NN	Type of specimen
collection_site	VARCHAR(100)		Body site specimen was collected from
collected_by	UUID	FK NN	→ users.user_id
collected_at	TIMESTAMP	NN	Collection timestamp
received_at	TIMESTAMP		Time specimen received in lab
status	ENUM	NN	ordered / collected / received / processing / completed / rejected
rejection_reason	TEXT		Reason if specimen was rejected

Radiology

radiology_reports

Imaging studies performed and radiologist reports. One row per imaging request.

Column	Data Type	Flags	Description
report_id	UUID	PK	Unique radiology report ID
request_id	UUID	FK NN	→ investigation_requests.request_id
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
modality	ENUM	NN	xray / ct / mri / ultrasound / fluoroscopy / mammography / other
body_part	VARCHAR(100)		Body region imaged
scheduled_at	TIMESTAMP		Scheduled imaging appointment time
performed_at	TIMESTAMP		Actual imaging performed timestamp
findings	TEXT		Radiographer/radiologist findings
impression	TEXT		Diagnostic impression/conclusion
image_reference	VARCHAR(255)		DICOM image file path or PACS reference
performed_by	UUID	FK NN	→ users.user_id — radiographer
reported_by	UUID	FK	→ users.user_id — radiologist (if separate)
status	ENUM	NN	scheduled / performed / reported / verified
reported_at	TIMESTAMP		Report submission timestamp

Billing & Cashier

bills

Master bill record per visit. Tracks totals, discounts, and payment status.

Column	Data Type	Flags	Description
bill_id	UUID	PK	Unique bill ID
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
total_amount	DECIMAL(12,2)	NN	Total bill amount before payments
discount_amount	DECIMAL(12,2)	NN	Total discounts/waivers applied
discount_reason	TEXT		Documented reason for discount
insurance_covered	DECIMAL(12,2)	NN	Amount covered by insurance
patient_payable	DECIMAL(12,2)	NN	Amount the patient must pay
amount_paid	DECIMAL(12,2)	NN	Total amount received
balance	DECIMAL(12,2)	NN	Remaining outstanding balance
status	ENUM	NN	open / partial / paid / waived / claimed
created_by	UUID	FK NN	→ users.user_id — cashier
created_at	TIMESTAMP	NN	Bill creation timestamp
updated_at	TIMESTAMP	NN	Last update timestamp

bill_items

Individual line items on a bill. One row per charge (consultation, lab, drug, etc).

Column	Data Type	Flags	Description
item_id	UUID	PK	Unique bill line item ID
bill_id	UUID	FK NN	→ bills.bill_id
visit_id	UUID	FK NN	→ visits.visit_id
item_type	ENUM	NN	consultation / lab / radiology / medication / ward / procedure / other
description	VARCHAR(255)	NN	Description of the charge
quantity	DECIMAL(8,2)	NN	Quantity (days, units, etc.)
unit_price	DECIMAL(10,2)	NN	Price per unit
total_price	DECIMAL(12,2)	NN	Computed total (quantity × unit_price)
reference_id	UUID		Optional FK to source record (request_id, prescription_id, etc.)
created_at	TIMESTAMP	NN	Line item creation timestamp

payments

Payment transactions against a bill. Multiple payments allowed per bill.

Column	Data Type	Flags	Description
payment_id	UUID	PK	Unique payment transaction ID
bill_id	UUID	FK NN	→ bills.bill_id
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
payment_method	ENUM	NN	cash / insurance / mobile_money / card / waiver
amount	DECIMAL(12,2)	NN	Payment amount
reference_number	VARCHAR(100)		Transaction or receipt reference
notes	TEXT		Cashier notes
received_by	UUID	FK NN	→ users.user_id — cashier
paid_at	TIMESTAMP	NN	Payment timestamp

insurance_claims

Insurance claim submissions and tracking per visit.

Column	Data Type	Flags	Description
claim_id	UUID	PK	Unique insurance claim ID
bill_id	UUID	FK NN	→ bills.bill_id
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
insurer_name	VARCHAR(150)	NN	Insurance company name
policy_number	VARCHAR(100)	NN	Patient policy number
claim_amount	DECIMAL(12,2)	NN	Amount claimed from insurer
approved_amount	DECIMAL(12,2)		Amount approved by insurer
status	ENUM	NN	submitted / pending / approved / partial / rejected
submission_reference	VARCHAR(100)		Insurer submission reference number
submitted_at	TIMESTAMP		Claim submission timestamp
resolved_at	TIMESTAMP		Date claim was approved or rejected
rejection_reason	TEXT		Reason if claim was rejected

Pharmacy

dispensing_records

Tracks every drug dispensed by pharmacy. One row per prescription item dispensed.

Column	Data Type	Flags	Description
dispensing_id	UUID	PK	Unique dispensing record ID
prescription_id	UUID	FK NN	→ prescriptions.prescription_id
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
drug_name	VARCHAR(200)	NN	Drug dispensed
batch_number	VARCHAR(50)		Drug batch/lot number
expiry_date	DATE		Drug expiry date
quantity_dispensed	DECIMAL(8,2)	NN	Quantity dispensed
unit	VARCHAR(20)	NN	Unit of measure e.g. tablets, ml
interaction_alert_acknowledged	BOOLEAN	NN	Pharmacist acknowledged any drug interaction alert
dispensed_by	UUID	FK NN	→ users.user_id — pharmacist
dispensed_at	TIMESTAMP	NN	Dispensing timestamp
billing_cleared	BOOLEAN	NN	Billing clearance confirmed before dispensing

drug_inventory_transactions

Full audit trail of every stock movement — restocks, dispensing deductions, adjustments, and write-offs.

Column	Data Type	Flags	Description
transaction_id	UUID	PK	Unique transaction ID
inventory_id	UUID	FK NN	→ drug_inventory.inventory_id — drug affected
transaction_type	ENUM	NN	restock / dispensing / adjustment / write_off / return
quantity_change	DECIMAL(10,2)	NN	Positive = stock added, Negative = stock removed
quantity_before	DECIMAL(10,2)	NN	Stock level before this transaction
quantity_after	DECIMAL(10,2)	NN	Stock level after this transaction
reference_id	UUID		Optional FK to dispensing_records.dispensing_id or purchase order
batch_number	VARCHAR(50)		Batch number if applicable (restocks)
expiry_date	DATE		Expiry date of the batch (restocks)
unit_cost	DECIMAL(10,2)		Cost per unit for this transaction (restocks)
notes	TEXT		Reason for adjustment or write-off
performed_by	UUID	FK NN	→ users.user_id — pharmacist or admin
created_at	TIMESTAMP	NN	Transaction timestamp

drug_inventory

Pharmacy drug stock. One row per drug product stocked.

Column	Data Type	Flags	Description
inventory_id	UUID	PK	Unique inventory item ID
drug_name	VARCHAR(200)	NN	Generic drug name
brand_name	VARCHAR(200)		Brand/trade name
drug_code	VARCHAR(50)	UQ	Internal drug code
category	VARCHAR(100)		Drug category e.g. antibiotic, analgesic
unit	VARCHAR(20)	NN	Dispensing unit e.g. tablets, vials
quantity_in_stock	DECIMAL(10,2)	NN	Current stock quantity
reorder_level	DECIMAL(10,2)	NN	Stock level that triggers a low-stock alert
unit_cost	DECIMAL(10,2)		Purchase cost per unit
unit_price	DECIMAL(10,2)		Selling price per unit
location	VARCHAR(100)		Physical storage location in pharmacy
last_restocked_at	TIMESTAMP		Last restock timestamp
updated_at	TIMESTAMP	NN	Last inventory update timestamp

Ward / Inpatient

beds

Hospital bed registry. One row per physical bed.

Column	Data Type	Flags	Description
bed_id	UUID	PK	Unique bed ID
ward_name	VARCHAR(100)	NN	Ward or unit name
bed_number	VARCHAR(20)	NN	Bed identifier within the ward
bed_type	ENUM	NN	general / icu / hdu / isolation / maternity / paediatric / other
is_available	BOOLEAN	NN	Whether bed is currently unoccupied
is_active	BOOLEAN	NN	Whether bed is in service
notes	TEXT		Any notes about the bed

admissions

Formal inpatient admission record. Links visit to a bed and tracks length of stay.

Column	Data Type	Flags	Description
admission_id	UUID	PK	Unique admission record ID
visit_id	UUID	FK NN	→ visits.visit_id
patient_id	UUID	FK NN	→ patients.patient_id
bed_id	UUID	FK NN	→ beds.bed_id
admitting_doctor_id	UUID	FK NN	→ users.user_id — admitting doctor
admitting_diagnosis	TEXT	NN	Diagnosis at time of admission
admission_date	TIMESTAMP	NN	Date and time of admission
discharge_date	TIMESTAMP		Date and time of discharge
length_of_stay_days	DECIMAL(6,1)		Computed length of stay in days
discharge_diagnosis	TEXT		Final diagnosis at discharge
discharge_instructions	TEXT		Instructions given at discharge
discharge_order_by	UUID	FK	→ users.user_id — discharging doctor
status	ENUM	NN	active / discharged / transferred / deceased

inpatient_orders

Doctor orders for admitted patients: medications, nursing, diet, investigations.

Column	Data Type	Flags	Description
order_id	UUID	PK	Unique inpatient order ID
admission_id	UUID	FK NN	→ admissions.admission_id

Column	Data Type	Flags	Description
patient_id	UUID	FK NN	→ patients.patient_id
order_type	ENUM	NN	medication / nursing / diet / investigation / procedure / other
order_detail	TEXT	NN	Full order description
frequency	VARCHAR(50)		How often the order is carried out
start_date	DATE		Order start date
end_date	DATE		Order end date (if applicable)
ordered_by	UUID	FK NN	→ users.user_id — doctor
status	ENUM	NN	active / completed / discontinued
ordered_at	TIMESTAMP	NN	Order creation timestamp

nursing_notes

Nursing observations, interventions and progress notes for admitted patients.

Column	Data Type	Flags	Description
note_id	UUID	PK	Unique nursing note ID
admission_id	UUID	FK NN	→ admissions.admission_id
patient_id	UUID	FK NN	→ patients.patient_id
note_type	ENUM	NN	observation / intervention / progress / medication_given / ward_round
note_text	TEXT	NN	Free-text note content
vitals_bp	VARCHAR(20)		BP reading in this note if recorded
vitals_temp	DECIMAL(4,1)		Temperature in this note if recorded
vitals_pulse	INT		Pulse in this note if recorded
vitals_spo2	DECIMAL(4,1)		SpO2 in this note if recorded
authored_by	UUID	FK NN	→ users.user_id — nurse
authored_at	TIMESTAMP	NN	Note creation timestamp

Administration

users

Staff user accounts for one hospital. Scoped entirely to this tenant.

Column	Data Type	Flags	Description
user_id	UUID	PK	Unique user account ID
username	VARCHAR(50)	UQ NN	Login username
email	VARCHAR(150)	UQ NN	Email for notifications
password_hash	VARCHAR(255)	NN	Bcrypt hashed password
full_name	VARCHAR(200)	NN	Staff full name
role	ENUM	NN	hospital_admin / receptionist / triage_nurse / doctor / lab_technician / radiographer / pharmacist / cashier
department	VARCHAR(100)		Department the user belongs to
phone	VARCHAR(20)		Staff contact number
is_active	BOOLEAN	NN	Whether account is active
force_password_change	BOOLEAN	NN	Force password change on next login
password_expires_at	TIMESTAMP		Password expiry timestamp (90-day policy)
last_login_at	TIMESTAMP		Most recent login timestamp
mfa_enabled	BOOLEAN	NN	Whether MFA is enabled (mandatory for admin)
mfa_secret	VARCHAR(100)		Encrypted MFA secret key
created_by	UUID	FK	→ users.user_id — admin who created this account
created_at	TIMESTAMP	NN	Account creation timestamp
updated_at	TIMESTAMP	NN	Last account update timestamp

departments

Hospital department configuration.

Column	Data Type	Flags	Description
department_id	UUID	PK	Unique department ID
department_name	VARCHAR(100)	NN	Department name
department_type	ENUM	NN	reception / triage / consultation / laboratory / radiology / pharmacy / ward / icu / billing / admin
head_user_id	UUID	FK	→ users.user_id — department head
is_active	BOOLEAN	NN	Whether department is active

fee_schedules

Hospital fee schedule for all billable services and items.

Column	Data Type	Flags	Description
fee_id	UUID	PK	Unique fee schedule item ID
item_name	VARCHAR(200)	NN	Name of the billable item/service
item_code	VARCHAR(50)	UQ	Internal billing code
item_type	ENUM	NN	consultation / lab / radiology / medication / procedure / ward / other
standard_price	DECIMAL(10,2)	NN	Standard charge for the item
insurance_price	DECIMAL(10,2)		Insurance-negotiated price if different
is_active	BOOLEAN	NN	Whether this fee is currently in use
effective_from	DATE	NN	Date this fee schedule takes effect
effective_to	DATE		End date (null = current)
created_at	TIMESTAMP	NN	Record creation timestamp

password_reset_tokens

Temporary tokens for password reset requests. Expires after 1 hour.

Column	Data Type	Flags	Description
token_id	UUID	PK	Unique token ID
user_id	UUID	FK NN	→ users.user_id — user who requested reset
token_hash	VARCHAR(255)	NN	Hashed reset token (never store plain text)
expires_at	TIMESTAMP	NN	Token expiry timestamp (1 hour from creation)
is_used	BOOLEAN	NN	Whether token has already been used
used_at	TIMESTAMP		Timestamp token was consumed
requested_ip	VARCHAR(45)		IP address of reset request
created_at	TIMESTAMP	NN	Token creation timestamp

refresh_tokens

JWT refresh tokens for secure session management. Enables forced logout when account is deactivated.

Column	Data Type	Flags	Description
token_id	UUID	PK	Unique refresh token ID
user_id	UUID	FK NN	→ users.user_id
token_hash	VARCHAR(255)	NN	Hashed refresh token
device_info	VARCHAR(255)		Browser/device info for display in active sessions list
ip_address	VARCHAR(45)		IP address when token was issued

Column	Data Type	Flags	Description
expires_at	TIMESTAMP	NN	Token expiry timestamp
is_revoked	BOOLEAN	NN	Whether token has been revoked (logout or forced)
revoked_at	TIMESTAMP		Timestamp of revocation
created_at	TIMESTAMP	NN	Token issue timestamp

notifications

In-system notifications for critical events: critical lab results, queue calls, low stock alerts, discharge orders.

Column	Data Type	Flags	Description
notification_id	UUID	PK	Unique notification ID
recipient_user_id	UUID	FK NN	→ users.user_id — intended recipient
notification_type	ENUM	NN	critical_result / queue_call / low_stock / discharge_order / system / prescription_ready
title	VARCHAR(200)	NN	Short notification title
message	TEXT	NN	Full notification message
reference_type	VARCHAR(50)		Type of related record e.g. lab_result, visit, prescription
reference_id	UUID		ID of the related record
is_read	BOOLEAN	NN	Whether notification has been read
read_at	TIMESTAMP		Timestamp notification was read
created_at	TIMESTAMP	NN	Notification creation timestamp

audit_logs

Tamper-proof record of every user action in this hospital tenant.

Column	Data Type	Flags	Description
log_id	UUID	PK	Unique audit log entry ID
user_id	UUID	FK NN	→ users.user_id
action	VARCHAR(100)	NN	Action performed e.g. CREATE, UPDATE, DELETE, LOGIN
table_name	VARCHAR(100)	NN	Database table affected
record_id	UUID		ID of the affected record
old_values	JSONB		Previous field values before change
new_values	JSONB		New field values after change
ip_address	VARCHAR(45)		User IP address at time of action
session_id	VARCHAR(100)		Session identifier
created_at	TIMESTAMP	NN	Audit log entry timestamp

Part 2 Master Database (System Owner)

One central database owned and managed by the system owner company. Stores hospital tenant registrations, subscriptions, invoices, and super admin accounts. Contains no patient data.

Tenant Management

tenants

Registry of all subscribing hospitals. Central record per hospital tenant.

Column	Data Type	Flags	Description
tenant_id	UUID	PK	Unique tenant (hospital) ID
hospital_name	VARCHAR(200)	NN	Official hospital name
country	VARCHAR(100)	NN	Country where hospital is located
city	VARCHAR(100)	NN	City
address	TEXT		Physical address
primary_contact_name	VARCHAR(200)	NN	Hospital primary contact person
primary_contact_email	VARCHAR(150)	NN	Primary contact email
primary_contact_phone	VARCHAR(20)	NN	Primary contact phone
billing_email	VARCHAR(150)	NN	Email for subscription invoices
timezone	VARCHAR(50)	NN	Tenant timezone e.g. Africa/Dar_es_Salaam
currency	VARCHAR(5)	NN	Currency code e.g. TZS, USD
date_format	VARCHAR(20)	NN	Preferred date format
logo_url	VARCHAR(255)		URL of hospital logo
db_connection_string	TEXT	NN	Encrypted connection string to tenant isolated database
data_region	VARCHAR(50)		Geographic region where tenant DB is hosted
status	ENUM	NN	trial / active / suspended / terminated
trial_ends_at	TIMESTAMP		End of free trial period
created_by	UUID	FK NN	→ super_admins.super_admin_id
created_at	TIMESTAMP	NN	Tenant creation timestamp
updated_at	TIMESTAMP	NN	Last update timestamp

Subscription & Billing

subscription_plans

Available subscription tiers (Basic, Standard, Premium) with pricing and limits.

Column	Data Type	Flags	Description
plan_id	UUID	PK	Unique plan ID
plan_name	VARCHAR(50)	NN	Plan name e.g. Basic, Standard, Premium
description	TEXT		Plan description
max_users	INT		Maximum user accounts allowed (null = unlimited)
max_patients	INT		Maximum patient records (null = unlimited)
storage_gb	INT	NN	Storage limit in GB
modules_included	JSONB	NN	JSON array of included module keys
monthly_price	DECIMAL(10,2)	NN	Monthly subscription price
annual_price	DECIMAL(10,2)	NN	Annual subscription price (discounted)
annual_discount_pct	DECIMAL(4,1)	NN	Annual discount percentage
uptime_sla_pct	DECIMAL(5,2)	NN	Guaranteed uptime SLA percentage
backup_frequency_hours	INT	NN	Backup frequency in hours
is_active	BOOLEAN	NN	Whether plan is available for new subscriptions
created_at	TIMESTAMP	NN	Plan creation timestamp

subscriptions

Active or historical subscription for each hospital tenant.

Column	Data Type	Flags	Description
subscription_id	UUID	PK	Unique subscription ID
tenant_id	UUID	FK NN	→ tenants.tenant_id
plan_id	UUID	FK NN	→ subscription_plans.plan_id
billing_cycle	ENUM	NN	monthly / annual
start_date	DATE	NN	Subscription start date
end_date	DATE	NN	Subscription end/expiry date
grace_period_days	INT	NN	Days after expiry before suspension (default 7)
auto_renew	BOOLEAN	NN	Whether subscription auto-renews
status	ENUM	NN	trial / active / grace / suspended / cancelled / terminated
suspended_at	TIMESTAMP		Timestamp of suspension
cancelled_at	TIMESTAMP		Timestamp of cancellation
cancellation_reason	TEXT		Reason for cancellation
created_at	TIMESTAMP	NN	Subscription creation timestamp

invoices

Subscription invoices generated per billing cycle for each hospital.

Column	Data Type	Flags	Description
invoice_id	UUID	PK	Unique invoice ID
tenant_id	UUID	FK NN	→ tenants.tenant_id
subscription_id	UUID	FK NN	→ subscriptions.subscription_id
invoice_number	VARCHAR(30)	UQ NN	Auto-generated invoice number
billing_period_start	DATE	NN	Start of billing period
billing_period_end	DATE	NN	End of billing period
plan_name	VARCHAR(50)	NN	Plan name at time of invoice
amount	DECIMAL(10,2)	NN	Invoice amount
currency	VARCHAR(5)	NN	Invoice currency
due_date	DATE	NN	Payment due date
status	ENUM	NN	unpaid / paid / overdue / void
issued_at	TIMESTAMP	NN	Invoice issue timestamp
paid_at	TIMESTAMP		Timestamp of payment receipt

saas_payments

Payment receipts from hospitals for their subscriptions.

Column	Data Type	Flags	Description
payment_id	UUID	PK	Unique SaaS payment ID
invoice_id	UUID	FK NN	→ invoices.invoice_id
tenant_id	UUID	FK NN	→ tenants.tenant_id
amount	DECIMAL(10,2)	NN	Payment amount received
currency	VARCHAR(5)	NN	Payment currency
payment_method	VARCHAR(50)	NN	Payment method e.g. bank transfer, card
reference_number	VARCHAR(100)		Bank or payment gateway reference
recorded_by	UUID	FK NN	→ super_admins.super_admin_id
receipt_sent_at	TIMESTAMP		Timestamp receipt was emailed to hospital
paid_at	TIMESTAMP	NN	Payment date

Super Admin

super_admins

System owner staff accounts. Completely separate from hospital user accounts.

Column	Data Type	Flags	Description
super_admin_id	UUID	PK	Unique super admin ID
username	VARCHAR(50)	UQ NN	Super admin login username
email	VARCHAR(150)	UQ NN	Email address
password_hash	VARCHAR(255)	NN	Bcrypt hashed password
full_name	VARCHAR(200)	NN	Full name
role	ENUM	NN	super_admin / billing_manager / support
mfa_secret	VARCHAR(100)	NN	MFA secret (mandatory for all super admins)
is_active	BOOLEAN	NN	Whether account is active
last_login_at	TIMESTAMP		Last login timestamp
created_at	TIMESTAMP	NN	Account creation timestamp

super_admin_audit_log

Tamper-proof log of all Super Admin actions across all tenants.

Column	Data Type	Flags	Description
log_id	UUID	PK	Unique audit log entry ID
super_admin_id	UUID	FK NN	→ super_admins.super_admin_id
action	VARCHAR(100)	NN	e.g. CREATE_TENANT, SUSPEND_TENANT, IMPERSONATE
tenant_id	UUID	FK	→ tenants.tenant_id (if applicable)
action_detail	JSONB		Detailed action parameters and changes
is_impersonation	BOOLEAN	NN	Whether this was an impersonation session
ip_address	VARCHAR(45)		Super admin IP address
created_at	TIMESTAMP	NN	Log entry timestamp

announcements

System maintenance notices and announcements broadcast to hospital tenants.

Column	Data Type	Flags	Description
announcement_id	UUID	PK	Unique announcement ID
title	VARCHAR(200)	NN	Announcement title
body	TEXT	NN	Full announcement content
audience	ENUM	NN	all / selected
target_tenant_ids	JSONB		JSON array of tenant IDs if audience is selected
publish_at	TIMESTAMP	NN	When to display the announcement
expires_at	TIMESTAMP		When announcement expires
created_by	UUID	FK NN	→ super_admins.super_admin_id
created_at	TIMESTAMP	NN	Creation timestamp

subscription_audit_log

Log of all subscription lifecycle events: plan changes, payments, suspensions, reactivations.

Column	Data Type	Flags	Description
event_id	UUID	PK	Unique subscription event ID
tenant_id	UUID	FK NN	→ tenants.tenant_id
subscription_id	UUID	FK	→ subscriptions.subscription_id
event_type	ENUM	NN	plan_created / plan_upgraded / plan_downgraded / payment_received / suspension / reactivation / termination / grace_period_started / cancellation_requested
actor_id	UUID		Super admin ID or null if automated
actor_type	ENUM	NN	super_admin / system / hospital_admin
reason	TEXT		Reason or notes for the event
created_at	TIMESTAMP	NN	Event timestamp

Part 3 Table Relations

The following table summarises all foreign key relationships across both databases.

From	To	Type	Description
visits.patient_id	patients.patient_id	Many-to-One	A patient can have many visits over time
visits.insurance_id	patient_insurance.insurance_id	Many-to-One	A visit may reference a patient's insurance policy
patient_insurance.patient_id	patients.patient_id	Many-to-One	A patient may have multiple insurance policies
queues.visit_id	visits.visit_id	Many-to-One	A visit generates multiple queue entries across departments
triage_assessments.visit_id	visits.visit_id	One-to-One	Each visit has one triage assessment
consultations.visit_id	visits.visit_id	One-to-One	Each visit has one consultation record
diagnoses.consultation_id	consultations.consultation_id	Many-to-One	A consultation can have multiple diagnoses (provisional, differential, final)
investigation_requests.consultation_id	consultations.consultation_id	Many-to-One	A consultation can generate multiple lab and radiology requests
prescriptions.consultation_id	consultations.consultation_id	Many-to-One	A consultation can have multiple prescription items
lab_results.request_id	investigation_requests.request_id	One-to-One	Each lab request produces one result record

From	To	Type	Description
<code>specimens.request_id</code>	<code>investigation_requests.request_id</code>	One-to-One	Each lab request has one specimen tracking record
<code>radiology_reports.request_id</code>	<code>investigation_requests.request_id</code>	One-to-One	Each radiology request produces one report
<code>bill_items.bill_id</code>	<code>bills.bill_id</code>	Many-to-One	A bill has multiple line items (consultation, lab, drugs, etc.)
<code>payments.bill_id</code>	<code>bills.bill_id</code>	Many-to-One	A bill can have multiple payment transactions
<code>insurance_claims.bill_id</code>	<code>bills.bill_id</code>	One-to-One	An insurance bill generates one claim
<code>dispensing_records.prescription_id</code>	<code>prescriptions.prescription_id</code>	One-to-One	Each prescription item has one dispensing record
<code>admissions.visit_id</code>	<code>visits.visit_id</code>	One-to-One	An inpatient visit has one admission record
<code>admissions.bed_id</code>	<code>beds.bed_id</code>	Many-to-One	A bed can be occupied by multiple admissions over time
<code>inpatient_orders.admission_id</code>	<code>admissions.admission_id</code>	Many-to-One	An admission can have many doctor orders
<code>nursing_notes.admission_id</code>	<code>admissions.admission_id</code>	Many-to-One	An admission can have many nursing notes
<code>drug_inventory_transactions.inventory_id</code>	<code>drug_inventory.inventory_id</code>	Many-to-One	Every stock change is recorded as a transaction against a drug inventory item

From	To	Type	Description
<code>password_reset_tokens.user_id</code>	<code>users.user_id</code>	Many-to-One	A user can have multiple reset token requests over time (old ones expire)
<code>refresh_tokens.user_id</code>	<code>users.user_id</code>	Many-to-One	A user can have multiple active sessions from different devices
<code>notifications.recipient_user_id</code>	<code>users.user_id</code>	Many-to-One	A user can have many notifications
<code>subscriptions.tenant_id</code>	<code>tenants.tenant_id</code>	Many-to-One	A tenant can have a history of subscriptions
<code>invoices.tenant_id</code>	<code>tenants.tenant_id</code>	Many-to-One	A tenant has many invoices over time
<code>invoices.subscription_id</code>	<code>subscriptions.subscription_id</code>	Many-to-One	Each subscription cycle generates invoices
<code>saas_payments.invoice_id</code>	<code>invoices.invoice_id</code>	One-to-One	Each invoice has one payment record
<code>subscription_audit_log.tenant_id</code>	<code>tenants.tenant_id</code>	Many-to-One	All subscription events for a tenant are logged here